

D. ball reaches is momentarily at rest and at max height after first bounce

- 4 B** t = time taken for 1st chestnut to fall through 10 m
 t_1 = time taken for 1st chestnut to fall through first 1.5 m
 t_2 = time taken for 1st chestnut to fall through last 8.5 m
= time taken for 2nd chestnut to fall through 10 m

$$10 = \frac{1}{2} (9.81) t^2 \quad \rightarrow \quad t = 1.428 \text{ s}$$

$$1.5 = \frac{1}{2} (9.81) t_1^2 \quad \rightarrow \quad t_1 = 0.553 \text{ s}$$

$$t_2 = t - t_1 = 0.875 \text{ s}$$

Let u be the initial speed of 2nd chestnut.

$$10 = u (0.875) + \frac{1}{2} (9.81)(0.875)^2 \rightarrow u = 7.1 \text{ m s}^{-1}$$